Helping Hand

Preliminary Report: Wearable-Assisted ASL Learning

Brian Paz, Gael Garcia, Kali Schuchhardt, Srinitha Srikanth
Computer Engineering, CEN4908C
October 28, 2026
Evidence cutoff: October 27, 2026

Abstract

Helping Hand is a wearable-assisted American Sign Language (ASL) learning prototype intended to support beginner practice through sensor-based feedback and persistent learner progress. Five resistive flex sensors and a wrist-mounted accelerometer/gyroscope connect to an ESP32-S3, which streams measurements and static predictions over Bluetooth Low Energy to a Flutter Android application. The application provides account access, target-specific practice, recording and export tools, and local-first progress storage with Firebase synchronization. This report describes the design, scientific background, preliminary results, engineering constraints, community considerations, and remaining development. The original 36-class static classifier is preserved as a baseline; current firmware packages a separate 34-class model trained from recorded poses. For isolated words, a temporal convolutional network (TCN) was compared with CNN-GRU and CNN-LSTM alternatives using fifteen real recordings of Hello, Please, and Yes. All candidates correctly classified three validation trials; the TCN was selected by the predefined model-size tie-break at 48,324 bytes and subsequently classified three reserved test trials correctly. These small, neutral-orientation results do not establish performance across users or live conditions. Software checks and Android recorded replay support integration, while current physical validation remains incomplete. Approximately two damaged flex sensors have delayed reliable new data collection. Replacement sensors and a nine-axis IMU are ordered, and ten piezo buzzers have been acquired for planned finger-specific and directional feedback. Near-term work addresses sensor repair, calibration, PCB development, recognition robustness, haptic cue mapping, and reproducible end-to-end testing. Educational effectiveness and reliable corrective feedback remain objectives requiring user evaluation.

1 Introduction

Learning ASL requires practice of handshape, movement, orientation, location and nonmanual features. ASL is a natural language with grammar distinct from English; it is not a universal sign system [1]. Books and ordinary prerecorded examples provide references but cannot directly measure a learner's own glove-sensed movements. Helping Hand explores whether a portable sensor glove and companion application can make independent beginner practice more informative between interactions with instructors and fluent signers.

The original design identified hearing beginners as intended learners and proposed tactile correction, modular hardware and structured practice [2]. The current prototype implements sensing, wireless telemetry, static and limited word recognition, and saved progress. Corrective haptics remain under development. A recent study involving ASL educators identifies structured signing feedback as a useful technology direction while emphasizing pedagogical and cultural considerations [3]. Helping Hand therefore targets a bounded learning aid; recognition of selected signs alone cannot demonstrate ASL competence or improved learning.

2 Background and Technical Basis

A static classifier maps a single sensor vector to a label. Moving signs require temporal structure: the order and duration of observations can distinguish gestures that share individual poses. Lee, Chong and Chung demonstrated word-level ASL recognition using six IMUs and an LSTM-based system [4]. That study supports

feasibility, but its fingertip sensing, dataset and evaluation differ from Helping Hand's five flex sensors and single wrist IMU. Published accuracy is not transferable to this prototype.

TCNs use temporal convolutions to combine neighboring observations; dilation extends the temporal context without a recurrent state. Bai, Kolter and Koltun provide a general comparison of convolutional and recurrent sequence models in an author preprint [5]. Helping Hand compares TCN, CNN-GRU and CNN-LSTM on identical recordings and preprocessing rather than assuming an architecture is superior. Trial boundaries must remain intact during splitting: neighboring frames from one sign are correlated and should not appear as independent training and test examples.

Orientation is both a sensing problem and meaningful sign information. Quaternion-based activity-recognition research motivates evaluating frame transformations under sensor rotation [6]; it does not prove that removing orientation helps ASL. Accelerometer/gyroscope fusion can estimate tilt and short-term rotation, while a calibrated magnetometer can provide a heading reference. Kok and Schön demonstrate the importance of magnetic calibration and axis alignment [7]. The planned nine-axis upgrade will test this possibility; no orientation normalization is implemented in the current word demo. Haptic perception also requires testing during movement: Tajdari, Forsyth and Lim found lower vibrotactile identification accuracy during repetitive hand motion [8]. Their results motivate a glove-specific cue study, not a claim that our acquired buzzers provide effective guidance.

3 System Design

3.1 Acquisition, communication and inference

The glove connects five flex channels to ESP32-S3 analog inputs and reads its current accelerometer/gyroscope over I^2C. Firmware targets a 40 Hz stream and records sequence numbers and device timestamps for delivery checks. BLE notifications carry raw flex values, acceleration, angular rate, sensor status and static predictions to one shared Flutter connection service. The Nordic UART-style custom GATT service provides device-to-app notifications and an app-to-device write characteristic [9, 10]. Its text packet contract is project-specific. Firmware configures a 517-byte MTU and the app requests 512; negotiated delivery and sustained packet integrity still require target-phone measurements.

The static path runs a multilayer perceptron on ESP32. Current firmware defaults to a 34-class recorded model covering digits and letters except motion-based J/Z; the original synthetic 36-class baseline remains separately available. The recorded model uses five calibrated flex inputs, two hidden layers of 128 and 64 units, and a softmax output [11].

The word path runs locally on the phone. Each complete attempt is linearly resampled using the device clock, center-trimmed or edge-padded to 96 samples, and standardized using training-only statistics. Its eleven features are five raw flex readings followed by three acceleration and three angular-rate channels. A 16-filter projection feeds three residual TCN blocks with two kernel-five causal convolutions per block and dilations 1, 2 and 4. Temporal averaging, a 16-unit dense layer and softmax yield three word scores. Dropout and L2 regularization are already present [12].

3.2 Learner interaction and persistence

Learners sign in, choose Alphabet, Numbers or Words, and open a dedicated practice page. Connection state, predictions, confidence and retry feedback are visible. Static completion requires a matching target at at least 80% confidence for 750 ms and five matching packets, with gaps no greater than 250 ms. Word practice instead uses explicit Start attempt and Finish sign boundaries; completion requires the actual highest-scoring word to match the target at 80% confidence. Invalid packets, interruptions and cancellation discard incomplete attempts [13].

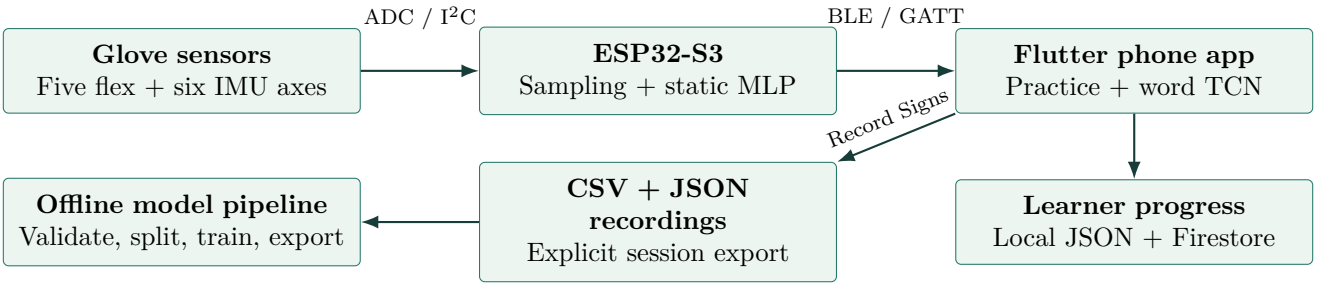


Figure 1: Implemented processing paths. Word inference is on the phone; training is offline. Proposed piezo feedback and the ordered nine-axis IMU are not shown as integrated components.

Progress is saved locally under the user's identity, then synchronized to an owner-scoped Firestore document. Firebase handles email/password authentication; raw recordings follow a separate app-private CSV/JSON export workflow [14, 15]. Offline local storage supports continued progress access, while live account isolation, word synchronization and recovery remain beta validation tasks.

3.3 Planned haptics and risk mitigation

Ten acquired piezo buzzers are planned: one per finger and five in an X on the back of the hand. They are intended to identify fingers requiring correction and communicate direction. Exact parts, driver requirements, pulse patterns and error-to-cue mapping remain unresolved. A word label does not identify which finger should move; a separate correction method must be designed. Initial work will validate one cue and actuator before expanding the array, checking comfort, distinguishability, power and sensor interference.

Two damaged flex sensors currently block reliable fresh acquisition. Replacement mounting will include contact strain relief, supported by Adafruit's handling guidance [16]. Recalibration and new recordings will address distribution changes. Further risks include confident predictions for unrelated motion, truncated long gestures, background completion of static lessons, and firmware waiting indefinitely for USB serial at startup. Existing records identify these issues; rejection testing, explicit practice lifecycle handling and standalone boot tests are planned mitigations [17].

4 Tools, Constraints and Engineering Standards

PlatformIO with Arduino/C++ builds ESP32 firmware; Flutter/Dart implements the Android interface [18, 19]. TensorFlow/Keras trains the models, with TFLite deployment and training-only scaling; Python tests and Flutter unit/widget/integration checks exercise the pipeline [20, 21, 22, 23, 24]. KiCad supports circuit design and board-layout planning, while Git records source revisions [25, 26]. These are development tools and processes, distinct from communication standards.

I²C supplies the sensor bus; current firmware configures 100 kHz, corresponding to Standard-mode's nominal maximum [27]. BLE/GATT defines wireless access to characteristics, with a custom UART-style service layered above it [9, 10]. The ordered Adafruit ICM-20948 breakout adds an AK09916 magnetometer and board-level regulation/level shifting. It requires different driver code and verification of axes, units, calibration and fusion before providing useful orientation features [28].

Constraints include limited independent recordings, sensor fragility, glove fit, battery capacity, actuator wiring and embedded memory. The recorded static firmware build uses 37.5% of configured RAM and 83.9% of application flash; the word model is deployed on the phone [11]. Listed replacement prices are $7.95 per flex sensor and $19.95 for the IMU as checked October 27; actual expenditure, full bill of materials and battery runtime remain unconfirmed [16, 28]. No product certification or measured power budget is claimed.

5 Needs, Impact and Professional Considerations

The initial need remains useful independent practice for hearing beginners, with instructor interaction and authentic ASL examples central to learning. The engineering contribution is integration of wearable sensing, local recognition, recording and account-specific practice, with proposed tactile guidance. A remote Notre Dame ASL-club discussion encouraged beginner progression, calibration and haptic feedback and noted the missing second hand and facial information. The participants were ASL learners reviewing media; their comments do not establish worn comfort, linguistic validation or representative user acceptance [29].

Cultural and social impact. Incorrect automatic feedback could reinforce errors or frustrate learners. Instruction and vocabulary should be reviewed by fluent signers and Deaf educators, including variation in signing and handedness. Deaf-led research identifies biases in sign-language AI agendas and datasets, supporting meaningful community participation [30]. A single glove cannot observe facial grammar, the uninstrumented hand or body-relative hand location reliably. Proposed learning benefits remain unmeasured.

Global impact. The hardware/software pattern may be reusable, but ASL does not represent all sign languages. Other languages require their own linguistic review, vocabulary, recordings and evaluation [1].

Economic and environmental impact. Requiring a glove and compatible phone may limit access. Replacement cost and downtime are practical constraints, and low purchase prices alone do not establish affordability. Replaceable sensors could extend service life; failed sensors, wiring and batteries still create waste. Repairability and appropriate disposal are relevant to the wider electronics-waste problem, but no life-cycle benefit has been measured [31].

Ethical and professional practice. Collection protocols call for consent, pseudonymous signer identifiers and controlled export of recordings. Account credentials belong to Firebase Authentication rather than sensor files. Actual sharing permissions and account isolation need review/testing. Reporting tiny datasets honestly, preserving failures, checking wearable electrical and mechanical limits, and crediting sources follow professional obligations to privacy, public welfare and realistic technical claims [32]. Future work should evaluate task usability before making educational-effectiveness claims; improved recognition alone does not demonstrate learning or retention.

6 Results and Evidence Limits

6.1 Alpha software and recording evidence

At commit `cce9d38` on September 10, project records report successful Flutter analysis, 16 Flutter tests, Android debug compilation, ESP32-S3 compilation and 23 backend tests [17]. These checks exercised software correctness and buildability. Historical videos show physical flex telemetry reaching an Android phone and changing its visualization, but do not provide a controlled accuracy study. The original physical location and complete device/run metadata are not recorded, limiting exact reproduction of that demonstration.

The September 17 static pilot saved 35 trials and 5,518 valid frames from one signer at neutral orientation. Record Signs retained raw packets, labels, device timestamps and session metadata; export review checked validity, ordering and sequence gaps. Documented rates were approximately 39.7–40.5 Hz, with no observed sequence gaps [33]. This supports recording/export feasibility under that session, not model accuracy; most labels have one trial.

6.2 Recorded static and word evaluation

The October 6 recorded static model used 52 whole trials, with seed 42 assigning 44 to training and four each to validation/test. Only A, B, 1 and 3 had held-out trials; the other 30 classes remained training-only.

Weighted training used Adam, batch size 32, an 80-epoch cap and validation-loss patience 8. The restored checkpoint correctly classified all 615 test frames and four trial-averaged predictions. Retraining reused the split during calibration revision, making this a regression check, not fresh untouched-test or all-class evidence. Calibration also reduced overall training fit and left errors including C/X, E/A and W/6 confusions [11]. This motivates fresh post-repair collection rather than relying on priority-sign success. Reproduction uses the two retained static exports, replaces older A/B/1/3 trials with the newer session, applies fourfold priority class weighting, and fits the scaler on training groups only. The recorded environment was Windows CPU with TensorFlow 2.21.0 and Keras 3.14.0; calibration, source hashes and frozen trial IDs are retained in the versioned record.

On October 5, the word comparison used fifteen trials, five per word, with seed 7 assigning three training, one validation and one test trial per word. All candidates used identical train-only scaling, 96-sample windows, 16-filter/ unit configurations, dropout 0.25, L2 0.0001, Adam learning rate 0.001, batch size 3, a 150-epoch cap and patience 20. Selection ranked validation accuracy, macro F1, file size and desktop latency, in that order. Only the selected candidate was then evaluated on reserved tests [12].

Candidate	Validation	Macro F1	TFLite bytes	Median ms
TCN	3/3	1.00	48,324	0.0488
CNN-GRU	3/3	1.00	180,212	0.0442
CNN-LSTM	3/3	1.00	159,320	0.0611

Table 1: Same-split word comparison on three validation trials. Desktop CPU timings exclude capture, BLE, preprocessing and UI; they are not phone latency.

The expected outcome was a compatible, compact candidate, not a demonstrated accuracy advantage. TCN won the size tie-break and correctly classified all three reserved tests, yielding macro F1 1.00 and one correct trial per class in the confusion matrix. Top-five accuracy is inapplicable to three classes. These neutral recordings came from the reported demo wearer; metadata signer IDs do not establish independent users. Desktop timing used one TFLite thread, ten warmups and 100 measured copy/invoke/output-copy calls. Source exports, split assignments, preprocessing and training procedures are identified in the training record. The input is the retained word CSV/manifest pair; reserved trials are `word_hello_0008`, `word_please_0011` and `word_yes_0005`. Evaluation compares the highest-scoring label with each trial's label, then computes class F1 and the confusion matrix. Device timestamps supply resampling because repeated phone receive times were flagged. Full generated runs require the team's retained artifact backup [12].

6.3 Beta integration and outstanding tests

October software integration checks covered Python/Dart input parity, completion/cancellation, progress and widgets: 42 focused checks passed. Three native recorded replays passed on a Pixel 8 API 36 emulator using the packaged TCN, with predictions within 0.02 percentage points of Python; the ordinary Android debug APK also built [13]. Replay formed complete attempts from the already evaluated recordings, ran the packaged interpreter and compared its output with the Python reference. The expected matching predictions were obtained without a physical glove or Firebase sign-in. It verifies deployment consistency, not additional held-out accuracy.

Fresh glove/phone recognition, sustained BLE delivery, intended-phone latency, current cloud word synchronization and external-user task tests remain **Not run**. Two broken flex sensors explain the current collection block. The T1/T2 procedures require recording build, device/OS, firmware, hardware, date, tester, expected result, observed measurements and evidence for future runs [34]. No new physical beta outcome is inferred from software checks or presentation delivery.

7 Current Status and Completion Plan

Account screens, practice routes, progress storage, recording/export, the static classifier and three-word TCN integration are implemented. Current hardware revalidation and instructional-media review remain incomplete. Since the October 6 demonstration, circuit design and board-layout planning have continued in KiCad; October 22 schematic files document progress, while saved board files contain no placed/routed layout. The current TCN review prioritizes broader data, complete-gesture windows, unrelated-motion rejection and sensor/orientation robustness. It establishes improvement directions, not measured gains [17, 35].

Date/window	State	Milestone / dependency
September 10–17	Recorded	Alpha builds/tests; real static recording/export pilot.
October 2–6	Implemented	Account/practice refinements; recorded model training and phone integration; live validation incomplete.
October 22–27	In progress	KiCad schematic/layout planning, TCN review; flex repair awaited; ten piezo buzzers acquired.
Next two weeks	Planned	Replace/calibrate flex sensors; integrate ICM-20948; complete circuit/driver design and initial haptic cues. Sequence depends on delivery and bench checks.
Before November 19	Proposed target	Collect fresh trials, evaluate revised features/model, verify live practice and progress, and test cue perception.
November 19	Scheduled	Post-mortem presentation.
December 3	Tentative	Evening CpE Showcase, approximately 6 p.m., WERT Lobby.

Table 2: Completed work and team-reported roadmap as of October 27. Future engineering dates are targets; the final written-report deadline is unconfirmed.

Repair will establish per-finger ranges before new data collection. IMU work will verify all nine channels and compare calibrated fusion against a baseline, preserving meaningful palm orientation. New input features require matching training and phone preprocessing. Haptic work must connect a defined correction signal to bounded actuator commands and validate perception with sensors monitored both during and after activation. Final checks should include wrong signs, idle motion, disconnects, restart, account isolation and repeated latency measurements. Independent trials and held-out users are needed before broader recognition claims. The remaining piezo product link and driver specifications are outstanding design inputs, not assumed specifications.

8 Conclusion

Helping Hand has progressed from static sensing and telemetry to an integrated practice application with recorded-data pipelines and a compact three-word TCN. Existing software and offline results support this preliminary prototype while leaving live reliability, generalization and learning benefits unresolved. Damaged flex sensors have delayed acquisition, but circuit planning and model review have continued. Replacement sensors, the ICM-20948 and the acquired piezo array provide a concrete next phase: calibrate the repaired glove, evaluate recognition changes, implement interpretable correction cues and record reproducible system/user tests. The project can provide a foundation for broader vocabulary and feedback research when these evidence gaps are addressed and ASL community expertise informs subsequent development.

References

[1] NIDCD, "American Sign Language." https://www.nidcd.nih.gov/health/american-sign-language. Accessed Oct. 27, 2026.

[2] Helping Hand team, *Helping Hand Design Draft*, first-semester design report. Project design archive, 2026.

[3] S. Hassan et al., "ASL Educators' Perspectives on AI for Enhancing Student Learning in American Sign Language Education," *CHI*, 2026, 20 pp. doi:10.1145/3772318.3791928.

[4] B. G. Lee, T.-W. Chong and W.-Y. Chung, "Sensor Fusion of Motion-Based Sign Language Interpretation with Deep Learning," *Sensors*, 20(21), 6256, 2020. doi:10.3390/s20216256.

[5] S. Bai, J. Z. Kolter and V. Koltun, "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling," author preprint, 2018. https://arxiv.org/abs/1803.01271.

[6] A. Yurtman, B. Barshan and B. Fidan, "Activity Recognition Invariant to Wearable Sensor Unit Orientation Using Differential Rotational Transformations Represented by Quaternions," *Sensors*, 18(8), 2725, 2018. doi:10.3390/s18082725.

[7] M. Kok and T. B. Schön, "Magnetometer Calibration Using Inertial Sensors," *IEEE Sensors Journal*, 16(14), pp. 5679–5689, 2016. doi:10.1109/JSEN.2016.2569160.

[8] M. Tajdari, J. Forsyth and S. Lim, "Sensitivity to Vibrotactile Stimulation in the Hand and Wrist: Effects of Motion, Temporal Patterns, and Biological Sex," *Human Factors*, online Aug. 23, 2024. doi:10.1177/00187208241275734.

[9] Bluetooth SIG, "The Bluetooth Low Energy Primer," ATT/GATT and notification sections. https://www.bluetooth.com/bluetooth-le-primer/.

[10] Nordic Semiconductor, "Nordic UART Service (NUS)." https://nrfconnectdocs.nordicsemi.com/ncs/latest/nrf/libraries/bluetooth/services/nus.html.

[11] Helping Hand team, *Recorded Static MLP for the October 6 Demo*, Oct. 6, 2026. Versioned training, calibration and verification record.

[12] Helping Hand team, *Recorded Word Demo Model*, Oct. 5, 2026. Versioned dataset, split, training and evaluation record.

[13] Helping Hand team, *TCN Word Practice Integration*, Oct. 5, 2026. Versioned integration and software-verification record.

[14] Firebase, "Authenticate with Firebase using password-based accounts on Flutter." https://firebase.google.com/docs/auth/flutter/password-auth.

[15] Firebase, "Writing conditions for Cloud Firestore Security Rules." https://firebase.google.com/docs/firestore/security/rules-conditions.

[16] Adafruit Industries, "Short Flex Sensor," product 1070, https://www.adafruit.com/product/1070; Spectra Symbol, *Flex Sensor Datasheet*, Rev. A2, https://cdn-shop.adafruit.com/datasheets/SpectraFlex2inch.pdf. Accessed Oct. 27, 2026.

[17] Helping Hand team, *Helping Hand Repository*: source, README, firmware configuration, recorded exports and KiCad files; snapshot a35d0f5, Oct. 27, 2026. https://github.com/GG1627/helping-hand/tree/a35d0f5.

[18] PlatformIO, "What is PlatformIO?" https://docs.platformio.org/en/latest/what-is-platformio.html.

[19] Flutter, "Flutter documentation." https://docs.flutter.dev/.

[20] TensorFlow, "Keras: The high-level API for TensorFlow." https://www.tensorflow.org/guide/keras.

[21] Google AI Edge, "LiteRT for Microcontrollers." https://developers.google.com/edge/litert/microcontrollers/overview.

[22] scikit-learn, "StandardScaler." https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html.

[23] pytest, "pytest documentation." https://docs.pytest.org/en/stable/.

[24] Flutter, "Testing Flutter apps." https://docs.flutter.dev/testing/overview.

[25] KiCad, *Getting Started in KiCad*, reference documentation. https://docs.kicad.org/9.0/en/getting_started_in_kicad/getting_started_in_kicad.html. Reference version does not identify the team's installed release.

[26] Git, "git documentation." https://git-scm.com/docs/git.

[27] NXP, *I2C-bus Specification and User Manual*, UM10204, Rev. 7.0, Oct. 1, 2021. https://www.nxp.com/docs/en/user-guide/UM10204.pdf.

[28] Adafruit Industries, "TDK InvenSense ICM-20948 9-DoF IMU," product 4554, https://www.adafruit.com/product/4554; *ICM-20948 Arduino Guide*, https://learn.adafruit.com/adafruit-tdk-invensense-icm-20948-9-dof-imu/arduino. Accessed Oct. 27, 2026.

[29] Helping Hand team, *Our Findings from the ASL Glove Demos*, remote Notre Dame ASL-club discussion notes, recorded date Sept. 18, 2026 with incomplete fields. Versioned formative-feedback notes.

[30] A. Desai et al., "Systemic Biases in Sign Language AI Research: A Deaf-Led Call to Reevaluate Research Agendas," *LREC-COLING SignLang Workshop*, pp. 54–65, 2024. https://aclanthology.org/2024.signlang-1.6/.

[31] ITU/UNITAR, *Global E-waste Monitor 2024*, 2024. Official publication record.

[32] IEEE, "IEEE Code of Ethics." https://ieee-cas.org/about/ieee-code-ethics.

[33] Helping Hand team, *Static ASL Pilot Recording*, Sept. 17, 2026. Raw exports and dataset description.

[34] Helping Hand team, *T1 Alpha Test Plan* and *T2 Beta Test Plan*, Sept.–Oct. 2026. T1 procedures; T2 procedures and run-record requirements.

[35] Helping Hand team with Codex assistance, *R3 TCN Improvement Review*, Oct. 27, 2026. Current code-review findings; no new experimental result.